WORKED SOLUTION

STEP II 2007, Question 4

Pure • Trigonometric Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q4

The key idea

Expand everything with the compound-angle formulae and divide by $\cos A \cos B$: the equation becomes a relation in $t = \tan A$ and $u = \tan B$ with a tu term, a t term, a u term and a constant. Such an expression factorises into two linear brackets exactly when "completing the rectangle" leaves nothing over, and the leftover turns out to be a multiple of $\beta^2 + \gamma^2 - \alpha^2$. The three equations then all have $\alpha = 2$, $\beta = \sqrt{3}$, $\gamma = 1$, which satisfy the condition; the only work is choosing A and B correctly.

Solution

The factorisation

By the compound-angle formulae,

$$\begin{aligned}&\alpha\sin(A-B) + \beta\cos(A+B) - \gamma\sin(A+B)\\ &\quad= \alpha(\sin A\cos B - \cos A\sin B) + \beta(\cos A\cos B - \sin A\sin B)\\ &\qquad - \gamma(\sin A\cos B + \cos A\sin B).\end{aligned}$$

Since $\cos A\cos B \neq 0$ we may divide by it. Writing $t = \tan A$, $u = \tan B$, the equation is equivalent to $P(t,u) = 0$, where

$$P(t,u) = \alpha(t-u) + \beta(1-tu) - \gamma(t+u) = -\beta tu + (\alpha-\gamma)t - (\alpha+\gamma)u + \beta.$$

Since $\beta \neq 0$, put

$$m = -\frac{\alpha+\gamma}{\beta}, \qquad n = \frac{\alpha-\gamma}{\beta}.$$

Then $-\beta(t-m)(u-n) = -\beta tu + \beta n\,t + \beta m\,u - \beta mn = -\beta tu + (\alpha-\gamma)t - (\alpha+\gamma)u + \frac{\alpha^2-\gamma^2}{\beta}$, which matches P except in the constant term. Hence, for all t and u,

$$P(t,u) = -\beta(t-m)(u-n) + \frac{\beta^2+\gamma^2-\alpha^2}{\beta}. \qquad (*)$$

If. When $\alpha^2 = \beta^2 + \gamma^2$ the constant in $(*)$ vanishes, so (as $\beta \neq 0$) the equation is $(\tan A - m)(\tan B - n) = 0$, with m and n depending only on α, β, γ.

Only if. Suppose the equation reduces to $(t - m')(u - n') = 0$ for constants m', n', so that $P(t, u) = 0$ holds exactly when $t = m'$ or $u = n'$. Write $c = (\beta^2 + \gamma^2 - \alpha^2)/\beta$ and suppose $c \neq 0$. Every pair (m', u) is then a solution, so $P(m', u) = -\beta(m' - m)(u - n) + c$ is zero for every u. If $m' \neq m$ this is a non-constant linear function of u, which is not always zero; if $m' = m$ it equals $c \neq 0$, which is never zero. Either way we have a contradiction, so $c = 0$, that is, $\alpha^2 = \beta^2 + \gamma^2$. (If "reduces to" is read as "P is a constant multiple of $(t - m)(u - n)$", comparing the tu coefficients forces the multiple to be $-\beta$, and then $(*)$ again forces $c = 0$.)

Ask yourself What happens in the parts below if $\cos x = 0$, where we are not allowed to divide?

Multiply $(*)$ back through by $\cos A \cos B$. When $\alpha^2 = \beta^2 + \gamma^2$ we get

$$\alpha \sin(A - B) + \beta \cos(A + B) - \gamma \sin(A + B) = -\beta(\sin A - m \cos A)(\sin B - n \cos B), \qquad (**)$$

and both sides are now just combinations of sines and cosines, so this identity holds for *all* A and B (both sides are continuous and agree whenever $\cos A \cos B \neq 0$). Using $(**)$ we never need to treat $\cos A = 0$ separately.

In all three parts $\alpha = 2$, $\beta = \sqrt{3}$, $\gamma = 1$, and $\alpha^2 = 4 = \beta^2 + \gamma^2$. So

$$m = -\frac{3}{\sqrt{3}} = -\sqrt{3}, \qquad n = \frac{1}{\sqrt{3}},$$

and the equation is $(\sin A + \sqrt{3}\cos A)(\sin B - \frac{1}{\sqrt{3}}\cos B) = 0$. Note that $\sin A + \sqrt{3}\cos A = 0$ forces $\cos A \neq 0$ (otherwise $\sin A = 0$ too), so it is the same as $\tan A = -\sqrt{3}$; likewise the second bracket vanishes exactly when $\tan B = \frac{1}{\sqrt{3}}$.

Part (i)

Take $A = x$ and $B = \frac{1}{4}\pi$, so that $A - B = x - \frac{1}{4}\pi$ and $A + B = x + \frac{1}{4}\pi$. Since $\tan \frac{1}{4}\pi = 1 \neq \frac{1}{\sqrt{3}}$, the second bracket is non-zero, so we need $\tan x = -\sqrt{3}$:

$$x = \tfrac{2}{3}\pi \quad \text{or} \quad x = \tfrac{5}{3}\pi.$$

Part (ii)

Now $A = x$ and $B = \frac{1}{6}\pi$. But $\tan \frac{1}{6}\pi = \frac{1}{\sqrt{3}} = n$, so the second bracket is $\sin \frac{1}{6}\pi - \frac{1}{\sqrt{3}}\cos \frac{1}{6}\pi = \frac{1}{2} - \frac{1}{2} = 0$. By $(**)$ the left-hand side minus the right-hand side is identically zero: the equation is an identity, and *every* x with $0 \leq x < 2\pi$ satisfies it.

Part (iii)

We need $A - B = x + \frac{1}{3}\pi$ and $A + B = 3x$. Adding and subtracting,

$$A = 2x + \tfrac{1}{6}\pi, \qquad B = x - \tfrac{1}{6}\pi.$$

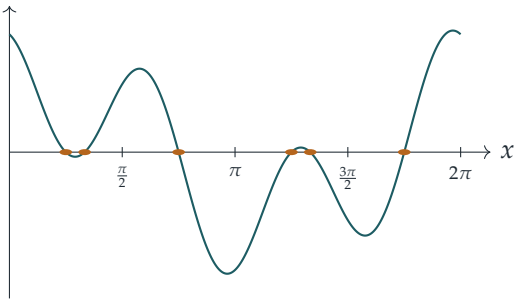
So the solutions are given by $\tan(2x + \frac{1}{6}\pi) = -\sqrt{3}$ or $\tan(x - \frac{1}{6}\pi) = \frac{1}{\sqrt{3}}$.

For the first, $2x + \frac{1}{6}\pi = \frac{2}{3}\pi + k\pi$, so $x = \frac{1}{4}\pi + \frac{1}{2}k\pi$ for integer k. In $[0, 2\pi)$ this gives $x = \frac{1}{4}\pi,\ \frac{3}{4}\pi,\ \frac{5}{4}\pi,\ \frac{7}{4}\pi$.

For the second, $x - \frac{1}{6}\pi = \frac{1}{6}\pi + k\pi$, so $x = \frac{1}{3}\pi + k\pi$, giving $x = \frac{1}{3}\pi,\ \frac{4}{3}\pi$.

These six values are distinct, so the solutions are

$$x = \tfrac{1}{4}\pi,\ \tfrac{1}{3}\pi,\ \tfrac{3}{4}\pi,\ \tfrac{5}{4}\pi,\ \tfrac{4}{3}\pi,\ \tfrac{7}{4}\pi.$$



The graph of the left-hand side minus the right-hand side of (iii) crosses zero exactly at the six marked points.

Checks

- In (i), at $x = \frac{2}{3}\pi$: $2\sin\frac{5}{12}\pi \approx 1.932$, $\sqrt{3}\cos\frac{11}{12}\pi \approx -1.673$, and $\sin\frac{11}{12}\pi \approx 0.259$; indeed $1.932 - 1.673 = 0.259$.
- In (ii), try $x = 0$: $2\sin(-\frac{1}{6}\pi) + \sqrt{3}\cos\frac{1}{6}\pi = -1 + \frac{3}{2} = \frac{1}{2} = \sin\frac{1}{6}\pi$. Any other x works too.
- In (iii), at $x = \frac{1}{4}\pi$: $2\sin\frac{7}{12}\pi \approx 1.932$ and $\sin\frac{3}{4}\pi - \sqrt{3}\cos\frac{3}{4}\pi \approx 0.707 + 1.225 = 1.932$. A numerical root search on $[0, 2\pi)$ finds exactly the six roots above.

Common slips

- Taking $A = x + \frac{1}{3}\pi$ and B from $3x$ in (iii): you must solve $A - B = x + \frac{1}{3}\pi$, $A + B = 3x$ simultaneously.
- Losing solutions in (iii) by stopping at $k = 0, 1$: the period of $\tan(2x + \frac{1}{6}\pi)$ is $\frac{1}{2}\pi$, so there are four solutions from that bracket.
- Missing that (ii) is an identity and hunting for particular solutions.

Beyond STEP

The expression $-\beta tu + (\alpha - \gamma)t - (\alpha + \gamma)u + \beta$ is a degenerate case of a conic: its zero set is a rectangular hyperbola unless a determinant-like quantity vanishes, when it splits into two lines. Deciding when a quadratic form factorises is a first step in classifying conics with matrices and eigenvalues, a topic in first-year linear algebra; see Strang, *Introduction to Linear Algebra*. Siklos, *Advanced Problems in Mathematics*, has more trigonometric equations in this spirit.

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